Lux 4.0 — GPU-Native Quasar Activation

QuasarSTM 4.0 execution + GPU-Residency invariant + 9-chain topology in production

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Spec freeze: February 7, 2026 | Activation: February 14, 2026

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Abstract

On 14 February 2026 the Lux Network activated its 4.0 epoch. Quasar consensus was promoted from version 3 to version 4, QuasarSTM execution graduated from version 3 to version 4 (LP-010 v4), the GPU-Residency invariant (LP-137) became mandatory for all canonical execution paths, and the 9-chain topology defined by LP-134 entered full production. The activation completes a four-year arc that began with mainnet launch in 2020, traversed a post-quantum upgrade in the 2.0 line, formalised the PQ suite at 3.0 (December 25, 2025), and now standardises GPU-native operation across every chain in the Lux family. This paper documents the launch as a retrospective: the chronology, the topology in production, the trifecta (D-Chain DEX, C-Chain EVM, F-Chain FHE) unifying every primary chain template, the white-label primary-chain resale to Liquidity, Zoo, Hanzo, and Pars, and the on-mainnet receipts captured between block heights $h_0 = 31{,}415{,}926$ and $h_0 + 10{,}000$.

Contents

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1 Chronology of the Lux Versions

Lux is now four major versions deep. Each version is a coherent release rather than a branch; each successor strictly subsumed its predecessor.

Version	Active from	Theme	Defining changes
1.0	Dec 2019 / 2020	Launch	Mainnet, P/C/X, snowball-derived consensus, native AMM on D-Chain (GPU).
2.0	2023–2024	PQ uplift	Hybrid signatures, Q-Chain bring-up, F-Chain alpha, lattice-based primitives in mempool.
3.0	2025-12-25	Full PQ + Quasar 3	ML-DSA mandatory, three-lane post-quantum certificates (LP-105), QuasarSTM 3, Quasar GPU FHE service alpha.
4.0	2026-02-14	GPU-native	Quasar 4, QuasarSTM 4, GPU-Residency invariant LP-137, 9-chain topology LP-134 in full production, white-label primary-chain templates.

Table 1: Lux major-version timeline. The D-Chain GPU AMM has been live since 1.0 — it predates the trifecta terminology by years and is the prior art the rest of the family inherits.

1.1 What 1.0 had that the family is now generalising

The D-Chain decentralised exchange (`lux.exchange`) shipped with the original 1.0 mainnet. It was GPU-native on day one: the AMM invariant solver, the price-router, and the order-book matching engine all ran on commodity GPUs in the validator pipeline. Quasar 4.0 makes that operating model the default *everywhere*.

2 The 9-Chain Topology in Production

LP-134 (*Canonical 9-chain Topology*) defines the production chain set. At h_0, every chain reported a non-zero block height with healthy validator participation.

Code	Chain	Purpose	4.0 status
P	Platform	Validators, staking, native asset	Quasar 4 cert lane 0x01
C	Contract (EVM)	GPU-native EVM, primary smart-contract surface	QuasarSTM 4 + cevm v0.41+
X	Exchange	UTXO asset transfers	Cert lane 0x02
Q	Quantum	Post-quantum signature service	Lane 0x03, ML-DSA primary
Z	ZK	ZK proof verification, privacy pool	ZKVM 4.0
A	Attestation	A-Chain attestation root (cf. `lux-achain-attestation`)	Bridges to Pars/Zoo/Hanzo
B	Bridge	Cross-chain bridge with 7-d fallback	MPC threshold lane
M	MPC	T-Chain threshold-signing service	FROST + Ringtail
F	FHE	GPU FHE service (cf. LP-013 v2)	CKKS / TFHE substrate

Table 2: Lux 4.0 9-chain topology. D-Chain was rolled into a sub-lane of C-Chain at the 4.0 boundary; the DEX precompile registry `0x0400–0x04FF` now lives on C-Chain so that DEX-using contracts pay no cross-chain hop.

The D-Chain absorption is the sole topological change at the 4.0 boundary; every other chain is a continuation.

3 Quasar 4.0 Consensus

Quasar 4.0 supersedes the three-lane post-quantum certificates of LP-105 with a four-lane variant (LP-020 v4):

1. **Lane 1 — BLS aggregate** (classical fast path, ≤ 100 ms).
2. **Lane 2 — ML-DSA aggregate** (post-quantum primary, ≤ 250 ms).
3. **Lane 3 — Ringtail threshold** (post-quantum threshold fallback, ≤ 800 ms).
4. **Lane 4 — SLH-DSA archival** (hash-only audit anchor, written once per epoch).

A block is *certified* when at least one of lanes 1–3 reaches $\geq 2/3$ stake-weight. Lane 4 produces the long-term tamper-evident record. The four-lane construction satisfies the safety property $\mathsf{Safe}_{4.0} \supseteq \mathsf{Safe}_{3.0}$: any 3.0-safe history is also 4.0-safe.

Theorem 3.1 (Live-fast / safe-slow)**.** *Under synchrony with $f < n/3$ Byzantine validators, Quasar 4.0 produces a certified block in time $T = \min(t_{BLS}, t_{ML\text{-}DSA})$ with probability 1 within the synchrony bound. Under partial synchrony, certification is guaranteed by the slower of the two lanes, and SLH-DSA continuity is preserved unconditionally.*

3.1 What changed from 3.0

- Lane addition (the SLH-DSA archival lane is new).
- Cert lane parallelism: lanes are signed concurrently across GPU threads rather than sequentially.
- Stake-weighted committee formation moved on-GPU (pre-4.0 it ran on the host CPU).
- Aggregation circuits are now resident on the GPU for the duration of an epoch.

4 QuasarSTM 4.0 Execution

LP-010 v4 specifies QuasarSTM 4.0; this paper does not duplicate the spec. Two operational consequences matter for the launch story:

Invariant 4.1 (GPU residency, LP-137)**.** *For every transaction τ executed at height $h \geq h_0$, the canonical execution path runs on a GPU device g, with all input state, intermediate state, and output state resident on g's HBM for the full duration of τ. Any host-RAM round-trip during τ is a violation.*

Invariant 4.2 (STM linearisability, LP-010 v4)**.** *Any execution of a block B under QuasarSTM 4.0 produces the same observable state and event log as a serial execution of B's transactions in the canonical order.*

The two invariants jointly guarantee that the GPU-resident execution is observationally indistinguishable from the textbook serial execution that 3.0 produced — only faster.

4.1 Production throughput at h_0

Metric	Lux 3.0	Lux 4.0 ($\mathbf{h_0}$)
C-Chain sustained TPS	24,000	360,000
D-Chain swap latency (median)	2.26 µs	0.94 µs
Block certification (median, lane 1)	620 ms	98 ms
ML-DSA-87 verifications/s/validator	11,000	620,000
Cross-chain bridge settlement	42 s	24 s
GPU-Residency violations / 24h	—	0 (target)

Table 3: Production throughput observed on Lux mainnet validators in the 24h following h_0.

The DEX figure is the long-running benchmark from `evmgpu-benchmark` updated to the 4.0 hardware target.

5 The Trifecta on Lux

Lux 4.0 is the reference implementation of the Lux-family trifecta: DEX + EVM + FHE. The trifecta is defined as the minimum financial-rail configuration of a real L1; networks lacking any of the three are toy.

5.1 DEX — D-Chain (lux.exchange), since 1.0

The D-Chain DEX is the prior art. Its precompile suite (`0x0400`–`0x04FF`) was promoted to C-Chain at the 4.0 boundary so that contract-resident DEX use no longer pays a cross-chain hop. Concrete features:

- Singleton pool manager (Uniswap-v4 style) at `0x0400`.
- HFT orderbook (LXBook) at `0x2000`.
- AMM pools (LXPool) at `0x2100`.
- Yield vaults (LXVault) at `0x2200`.
- Oracle feeds (LXFeed) at `0x2300`.
- Lending engine, transmuter, and perpetuals (`0x0410`–`0x0432`).
- Cross-chain hooks via Liquidity Protocol (LP-310, registered at $h_0 + 48$).

The D-Chain DEX has the longest production track record of any component in the family — six years of continuous operation at h_0 — and is the model the April 20 wave of network DEXes integrates against.

5.2 EVM — C-Chain (cevm v0.41+), GPU-native

C-Chain runs cevm v0.41 (cf. LP-009 *GPU-native EVM*) with the full canonical precompile registry:

- Standard EVM precompiles `0x01`–`0x0a`.
- Warp / Teleport (`0x0100`–`0x01FF`).
- Chain config (`0x0200`–`0x02FF`).

- AI/ML (`0x0300`–`0x03FF`).
- DEX (`0x0400`–`0x04FF`).
- Graph/Query (`0x0500`–`0x05FF`).
- Post-quantum (`0x0600`–`0x06FF`).
- Privacy (`0x0700`–`0x07FF`).
- Threshold signatures (`0x0800`–`0x08FF`).
- ZK proofs (`0x0900`–`0x09FF`).
- Curves (`0x0A00`–`0x0AFF`).

The 4.0 boundary added two: `Quasar` verification (0x0604) and `Hybrid_BLS_Ringtail` (0x0610), both GPU-native.

5.3 FHE — F-Chain, LP-013 v2

F-Chain hosts the GPU FHE service. The 4.0 boundary upgrades it from research-preview (3.0) to production:

- CKKS for approximate-arithmetic ciphertexts (price aggregation, risk computation).
- TFHE for boolean/integer ciphertexts (private voting, encrypted ACLs).
- Threshold decryption keyed by a t-of-n committee redrawn each epoch ($t = 67$, $n = 100$ on mainnet).
- Bootstrapping budget allocated as a first-class fee market; bootstrappers receive the new *F-Chain relay reward* (capped at 0.5% annualised LUX emission).

The GPU FHE service is the substrate that Hanzo, Zoo, and Pars re-export through their own primary chains.

6 White-Label Primary Chain Templates (LP-134)

LP-134 not only defines the canonical 9-chain topology; it also defines the *primary chain template*, the smallest reproducible configuration of P+C+D+F that constitutes a real L1. Four production deployments at h_0 run the template:

Deployment	Repository	Notes
Lux	`ghcr.io/luxfi/luxd`	Reference; the canonical instance.
Hanzo	`ghcr.io/hanzoai/hanzod`	4.0 launch 2026-02-14, AI-native.
Zoo	`ghcr.io/zooai/zood`	4.0 launch 2026-02-14, conservation-native.
Pars	`ghcr.io/luxfi/parsd`	2.0 launch 2026-02-14, diaspora-native.

Table 4: Production primary-chain deployments running the LP-134 template at h_0. Liquidity is a separate white-label deployment in its own jurisdictional silo and is not enumerated here.

Each deployment specialises by adding or replacing precompiles; the *topology*, *cert format*, *state model*, and *GPU residency requirement* are identical.

7 Activation Receipts

The 4.0 activation was block-aligned at $h_0 = 31{,}415{,}926$ on the canonical Lux main DAG.

Event	**Height**	**Notes**
Quasar 4.0 epoch boundary	h_0	cert-lane vector (BLS, MLDSA, Ringtail, SLHDSA)
QuasarSTM 4.0 activation	h_0	STM root rotated; all in-flight tx replayed
GPU-Residency invariant on	$h_0 + 1$	first violation counter = 0
SLH-DSA archival lane initial	$h_0 + 1$	first archival cert hash 0x91 ...
D-Chain rolled into C-Chain	$h_0 + 6$	DEX precompiles registered
Liquidity Protocol register	$h_0 + 48$	router id `lp.lux.router`
First Pars→Lux bridge tx	$h_0 + 218$	via A-Chain attestation
First Hanzo→Lux bridge tx	$h_0 + 305$	via Liquidity Protocol
First Zoo→Lux bridge tx	$h_0 + 412$	via Liquidity Protocol
F-Chain DKG threshold key	$h_0 + 720$	$t = 67$, $n = 100$

Table 5: Selected mainnet receipts from the Lux 4.0 activation.

7.1 Validator participation

Indicator	**Value at** $h_0 + 10\,000$
Validators voting	1,248
Stake-weighted participation	94.7%
Lane-1 (BLS) participation	94.3%
Lane-2 (ML-DSA) participation	93.8%
Lane-3 (Ringtail) participation	84.2%
Lane-4 (SLH-DSA archival) writes	100% of epochs

Table 6: Validator participation in the four-lane cert protocol.

8 Conclusion

Lux 4.0 is the activation of an operating model the network has been converging on since 2020: GPU-native, post-quantum, and trifecta-complete. The 4.0 boundary did not invent the trifecta; it formalised it. D-Chain has been the GPU-native DEX since 1.0; Quasar 4 and QuasarSTM 4 make every other chain operate the same way. The white-label primary chain template (LP-134) is the export mechanism: Hanzo, Zoo, and Pars all activate the same template on the same date, and the Liquidity Protocol common API on April 20 unifies their DEX liquidity.

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